HDOC-48

Pivot Index in an Integer Array

Aim

To write a C program that reads a variable-length integer array and finds the leftmost pivot index, where the sum of all elements to the left is equal to the sum of all elements to the right.

Algorithm

1. Read the size n of the array.
2. Read n integer elements into arr[].
3. Pass the array to pivot_index() using its address.
4. For each index i, calculate the left sum and right sum. For the first element, left sum is 0; for the last element, right sum is 0.
5. If left sum equals right sum, return i immediately so the leftmost pivot index is obtained.
6. If no such index is found, return -1.
7. Display the returned pivot index.

C Program

```
#include <stdio.h>

int pivot_index(int *arr, int n) {
   int i, j;
   int leftSum, rightSum;

   for (i = 0; i < n; i++) {
      leftSum = 0;
      rightSum = 0;

      for (j = 0; j < i; j++)
         leftSum += *(arr + j);

      for (j = i + 1; j < n; j++)
         rightSum += *(arr + j);

      if (leftSum == rightSum)
         return i;
   }
   return -1;
}

int main(void) {
   int n, i, result;
```

```
    printf("Enter the size of array: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Invalid array size.\n");
        return 0;
    }

    int arr[n];

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    result = pivot_index(arr, n);
    printf("Pivot index = %d\n", result);

    return 0;
}
```

Test Cases

Test Case	Input	Expected Output	Reason
1	n = 6 arr = [1, 7, 3, 6, 5, 6]	Pivot index = 3	Left sum = 11 and right sum = 11.
2	n = 3 arr = [2, 1, -1]	Pivot index = 0	Left sum = 0 and right sum = 0.
3	n = 3 arr = [1, 2, 3]	Pivot index = -1	No index has equal left and right sums.
4	n = 1 arr = [10]	Pivot index = 0	Both left sum and right sum are 0.

Sample Output

Enter the size of array: 6
Enter 6 elements: 1 7 3 6 5 6
Pivot index = 3

Result

The program successfully finds and displays the leftmost pivot index of the given integer array. If no pivot index exists, it displays -1.